

QUINLONG LIANG

Mechanical Engineering Student | Robotics and Electromechanical Prototyping

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Portfolio, videos and 3D models: quinwithonen.github.io/Longfly/quin

TECHNICAL PROFILE

Mechanical engineering student with hands-on experience designing, assembling, programming, and selling UAV hardware. Built VTOL aircraft, flight-controller systems, Arduino control projects, 3D-printed prototypes, and a magnetic levitation system. Available for full-time and contract work in Greater Boston.

TECHNICAL SKILLS

Design and prototyping: SolidWorks, 3D modeling, 3D printing, mechanical design, UAV assembly, prototype development

Programming and controls: C coursework, C++ and Arduino, PID control, sensor integration, embedded prototyping

Engineering: thermodynamics, fluid dynamics, statics, tolerance and center-of-gravity calculations, component selection, technical documentation

SELECTED TECHNICAL PROJECTS

Founder and Hardware Developer | Longfly.co

Feb 2024 - Present

- Designed and built a programmable UAV flight-controller platform; sold 13 units and distributed 60 total to Brown University's drone club, UMass Lowell, and independent users in the United States and abroad.
- Programmed VTOL flight controls in Arduino C++ using the open-source dRehmFlight project; added proportional-control and PID-gain readout functions for tuning and testing.
- Created SolidWorks models and 3D-printed VTOL airframes and PID test stands; specified Teensy 4.0 pinouts and mechanical interfaces while collaborating with an electrical engineer on a PCB incorporating a 5 V buck converter.

Magnetic Levitation Control System | Independent Project

- Designed, assembled, and programmed a one-axis magnetic levitator from scratch using an Arduino, electromagnet, and two Hall-effect sensors; wrote the control code to interpret sensor inputs and regulate magnet actuation.

Founder | Shanghai Jiao Tong University Drone Club

Jan 2026 - Present

- Founded and led a student drone club, recruited members, managed a CNY 10,000 department budget, and acquired five RC aircraft plus supporting components.
- Led the build of an RC F4U Corsair with a VR head-tracking FPV camera gimbal; built a VR-controlled aerial camera platform using a gimbal-mounted camera and a VR headset interface.

Team Lead | UML UAV Sustainability Competitions

Sep 2022 - Apr 2023

- Led an engineering and business team that won \$5,500 across two sustainability competitions for a UAV concept based on NASA PRANDTL-D geometry; designed a variable-lift-distribution wind-tunnel experiment.

ENGINEERING AND TECHNICAL EXPERIENCE

Product Engineering Intern | Freak Athlete Fitness

May 2023 - Oct 2023

- Evaluated manufacturing feasibility, material requirements, bushing tolerances, center-of-gravity placement, and expected production costs for fitness equipment; proposed folding mechanisms for the Nordic Hyper and Wheelbarrow Sled.
- Did CAD for the Wheelbarrow Sled around standard parts and stock sheet-metal sizes, designing for weldability and manufacturability.

Creator and Instructor | South End Technology Center Drone Curriculum

Nov 2020 - Apr 2021

- Secured approval and \$2,500 in funding, managed component procurement, and taught six students FPV-drone design, power management, protocol selection, hardware setup, and PID tuning.

International Sales Lead | M7 Cycling

Aug 2026 - Present

- Built an international sales pipeline and automated marketplace-listing workflow for a bicycle refurbishment business while conducting customer discovery with overseas bicycle shops.
- Generated \$7K in revenue within 10 days from the first three bikes sold abroad.

EDUCATION

University of Michigan | B.S. Mechanical Engineering | GPA 3.2

Expected Dec 2027

Shanghai Jiao Tong University | Mechanical Engineering Exchange (ranked #2 in China for ME)

2025 - 2026

University of Massachusetts Lowell | Mechanical Engineering Coursework

2022 - 2024

Coursework: C programming, SolidWorks, calculus, differential equations, thermodynamics, fluid dynamics, statics, and electricity and magnetism.